

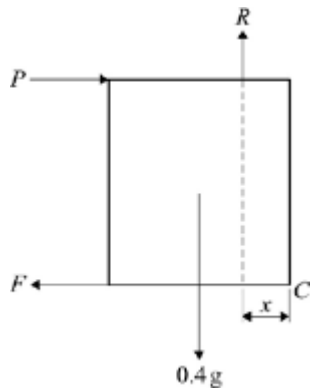
## Mark schemes

### Q1.

|     | Marking Instructions                                                                                  | AO     | Marks | Typical Solution                                                                                                                                                                                                                                                                |
|-----|-------------------------------------------------------------------------------------------------------|--------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Forms an integral to find mass.                                                                       | AO1.1a | M1    | $M = p \int_0^2 \frac{\pi x^4}{4} dx$ $= \pi \rho \left[ \frac{x^5}{20} \right]_0^2$ $= \frac{8}{5} \pi \rho$ $\bar{x} \times \frac{8}{5} \pi \rho = \int_0^2 \frac{\pi p x^5}{4} dx$ $= \pi p \left[ \frac{x^6}{24} \right]_0^2$ $= \frac{8}{3} \pi p$ $\bar{x} = \frac{5}{3}$ |
|     | Obtains correct mass.                                                                                 | AO1.1b | A1    |                                                                                                                                                                                                                                                                                 |
|     | Forms an integral of the form $\int xy^2 dx$                                                          | AO1.1a | M1    |                                                                                                                                                                                                                                                                                 |
|     | Obtains correct value of $\int xy^2 dx$                                                               | AO1.1b | A1    |                                                                                                                                                                                                                                                                                 |
|     | Obtains correct centre of mass, follow through 'their' mass provided both M1 marks have been awarded. | AO1.1b | A1F   |                                                                                                                                                                                                                                                                                 |
| (b) | Uses trigonometry and 'their' centre of mass to form equation to find angle.                          | AO3.1b | M1    | $\tan \alpha = \frac{2}{2 - \frac{5}{3}}$ $\alpha = 80.5^\circ$                                                                                                                                                                                                                 |
|     | Finds the correct angle to the required accuracy.                                                     | AO1.1b | A1F   |                                                                                                                                                                                                                                                                                 |
|     | FT 'their' values provided all M1 marks have been awarded                                             |        |       |                                                                                                                                                                                                                                                                                 |
|     |                                                                                                       |        |       | <b>Total 7 marks</b>                                                                                                                                                                                                                                                            |

### Q2.

On point of sliding:



$$\left. \begin{array}{l} P = F \\ R = 0.4g \\ F = \mu R \end{array} \right\} P = 0.4g \mu$$

Obtaining  $P$  in terms of  $\mu$

B1

Moments about C:

$$P(0.28) + Rx = 0.4g(0.10)$$

M1 Forming a moment equation at least one term correct

M1

$$(0.4g \mu)(0.28) + (0.4g)x = 0.4g(0.10)$$

A1 – two terms correct

A2 – all terms correct  $x, \mu$  only

A2,1

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$$x > 0 \Rightarrow x = 0.1 - 0.28\mu > 0$$

$$\mu < \frac{0.1}{0.28} = \frac{10}{28} = \frac{5}{14}$$

Using  $x > 0$

M1

$$k = \frac{5}{14} \text{ or } 0.357$$

CAO

A1

**Alternative**

$$\left. \begin{array}{l} F = \mu R \end{array} \right\} P = 3.92\mu$$

To slide:  $R = 0.4g$

$$F = p$$

Obtaining  $P$  in terms of  $\mu$

(B1)

Consider toppling:  $P$  needed

Take moments about  $C$  to find  $P$ :

$$0.4g(0.10) = P(0.28)$$

*M1 – One side correct, A1 all correct*

(M1A1)

$$1.4 = P$$

*A1 – Correct  $P$  value*

(A1)

$$P_{\text{sliding}} < P_{\text{toppling}}$$

$$\text{Slides before topples} \Rightarrow 3.92\mu < 1.4$$

*Inequality for  $\mu$  using  $P_{\text{sliding}} < P_{\text{toppling}}$*

(M1)

$$\mu < \frac{1.4}{3.92} = \frac{5}{14} = k$$

*Accept 0.357 (NB M0 if inequality reversed in equation)*

(A1)

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[6]

**Q3.**

(a)  $\pi \int xy^2 dx = \pi \int_0^{2r} x(x + 3r)^2 dx$   
*(use of  $\pi$  must be consistent)*  
*attempt to integrate  $\int xy^2 dx$*   
*- must involve three terms*

M1

$$= \pi \int_0^{2r} (x^3 + 6rx^2 + 9r^2x) dx$$

$$= \pi \left[ \frac{x^4}{4} + 2rx^3 + \frac{9r^2x^2}{2} \right]_0^{2r}$$

*correct integration*

A1

$$\pi[4r^4 + 16r^4 + 18r^4]$$

*Limits used correctly*

m1

$$38r^4\pi$$

correctly evaluated in terms of  $r^4$

A1

$$x = \frac{38r^4\pi}{98\pi r^3/3} = \frac{57r}{49}$$

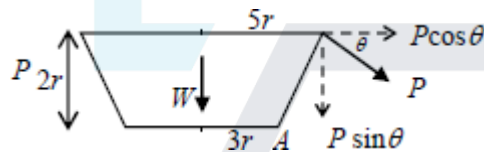
M1

$\frac{\pi \int xy^2 dx}{\text{use of volume}}$   
 ft 'their'  $\int xy^2 dx$

A1F

6

(b) (i)



Moments at A:

$$W(3r) = P \sin \theta (2r) + P \cos \theta (2r)$$

M1 attempt at moments - evidence of

force  $\times$  perpendicular distance

A1 two terms correct

M1A1

A1 all terms correct

A1

$$3W = 2P(\cos \theta + \sin \theta)$$

$$\frac{3W}{2(\cos \theta + \sin \theta)} = P$$

AG Must see evidence of factorising

A1

4

(ii) Min value of  $P$  is when  $\cos \theta + \sin \theta$  is at a maximum.



M1

Max value of  $\cos\theta + \sin\theta$  is  $\sqrt{2}$   
 $\sqrt{2}$  seen

A1

$$\text{Max } P \text{ value} = \frac{3W}{2\sqrt{2}}$$

$$\text{Or equiv eg } \frac{6\sqrt{2W}}{8}, \frac{3}{4}\sqrt{2}W \text{ etc}$$

A1

3

(iii)  $\theta = 45^\circ$

B1

1

[14]

Q4.

(a) Volume of cone  $= \frac{1}{3}\pi r^2 h = \frac{8h^3\pi}{3}$

Correct volume

B1

Alternative:  $\left[ \pi \int_0^{8h} \left( \frac{1}{8}x \right)^2 dx = \frac{8h^3}{3}\pi \right]$

(B1)

$$\pi \int xy^2 dx = \pi \int_0^{8h} x \left( \frac{x^2}{8} \right) dx = \pi \int_0^{8h} \frac{x^3}{64} dx$$

Use of  $\int xy^2 dx$

M1

$$= \pi \left[ \frac{x^4}{256} \right]_0^{8h}$$

Correct integration of  $kx^3$ 

A1

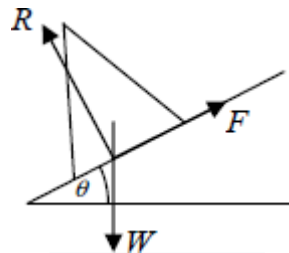
$$= 16h^4$$

Attempt at formula for  $\bar{x}$

$$= 6h$$

A1

(b) (i)



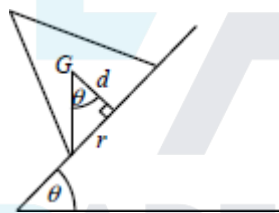
3 forces

*Through a common point at base of cone*

B1  
B1

2

(ii)



G above point of rotation

$$\tan \theta = \frac{r}{d}$$

*tan  $\theta$  seen anywhere*

M1

$$= \frac{h}{8h - 6h} = \frac{1}{2}$$

*Ratio correct*

A1

$$\theta = \tan^{-1} \frac{1}{2} = 26.6^\circ$$

Use of  $\tan^{-1}$  to get  $\theta$ ; ft incorrect ratio

A1F

3

(c)

$$\left. \begin{array}{l} \text{Resolve perpt to plane } N = W \cos \alpha \\ \text{Parallel to plane } F = W \sin \alpha \end{array} \right\}$$

Both correct

B1

Limiting friction  $F = \mu N \Rightarrow \tan \alpha = \frac{6}{13}$

Eliminate  $W$ , obtain  $\tan \alpha$

M1

$\alpha = 24.8^\circ < 26.6^\circ$  or  $\frac{6}{13} < \frac{1}{2}$

Comparison

A1

$\Rightarrow$  slides first

Conclusion; ft (b)(ii)

A1F

4

[14]

**Q5.**

(a) Moments about C:

$P \cos \theta$  seen

B1

$mga = P \cos \theta$

Forming moments equation – 2 terms

M1

$P = \frac{mg}{\cos \theta}$

A1

3

(b) Resolve  $\leftrightarrow$   $F = P \cos \theta$  1

Resolve in two directions

M1

Resolve  $\uparrow$   $mg = R + P \sin \theta$  2

Both equations correct

A1

Friction law (sliding)  $F = \mu R$  3

Substituting 1 and 2 in 3:

$P \cos \theta = \mu (mg - P \sin \theta)$

Substituting in  $F = \mu R$  – dep on first M1

m1

$$P \cos \theta + P\mu \sin \theta = \mu mg$$

$$P = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$$

AG

A1

4

(c) Slides first  $\Rightarrow$

$$\frac{\mu mg}{\cos \theta + \mu \sin \theta} < \frac{mg}{\cos \theta}$$

Set up inequality –  
expression in (b) < expression in (a)(ii)

$$\mu \cos \theta < \cos \theta + \mu \sin \theta$$

Correct simplification – remove fractions  
ft parts (a) and (b)

M1

$$\mu(\cos \theta - \sin \theta) < \cos \theta$$

$$\mu < \frac{\cos \theta}{\cos \theta - \sin \theta}$$

A1F

CAO ; Alternative:  $\mu < \frac{1}{1 - \tan \theta}$

A1

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3

(d) Inequality independent of mass, so no change

**No change (E1) and reason (E1)**  
ft error in (c); must give consistent reason  
If no reason, E0

E2,1F

2

[12]

**Q6.**

(a)  $\frac{1}{2} \int_{-2}^2 y^2 \, dx$

$$= \frac{1}{2} \int_{-2}^2 (16 - 8x^2 + x^4) \, dx$$

Attempt to integrate  $y^2$  as a function of  $x$



M1

$$= \frac{1}{2} \left[ 16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_{-2}^2$$

Correct integration

A1

$$= \frac{256}{15}$$

Correct limits applied to their integral

A1F

$$\bar{y} = \frac{256/15}{32/3}$$

$$\bar{y} = \frac{\frac{1}{2} \int_a^b y^2 dx}{\text{Area}}$$

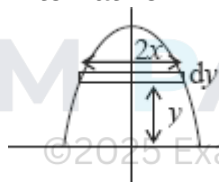
m1

$$= \frac{8}{5}$$

A1

5

Alternative 1:



$$\begin{aligned} I &= \int_{x=-2}^{x=0} 2xy \, dy = \int_2^0 2x(4-x^2)(-2x) \, dx \\ &= \int_2^0 4x^4 - 16x^2 \, dx \end{aligned}$$

Attempt to integrate  $2xy$  as a function of  $x$

(M1)

$$I = \left[ \frac{4x^5}{5} - \frac{16x^3}{3} \right]_2^0$$

Correct integration

(A1)

$$I = \frac{256}{15}$$

Correct limit applied to their integral

(A1F)

$$\bar{y} = \frac{I}{32/3} = \frac{8}{5}$$

Their evaluated  $I \div$  area

(m1)(A1)

**Alternative 2:**

$$I = \int_{y=0}^{y=4} 2xy \, dy = \int_0^4 2\sqrt{(4-y)}y \, dy$$

Attempt to integrate  $2xy$  as a function of  $y$   
by parts

(M1)

$$I = \left[ -\frac{4y}{3}(4-y)^{\frac{3}{2}} \right]_0^4 + \int_0^4 \frac{4}{3}(4-y)^{\frac{3}{2}} \, dy$$

$$I = \left[ -\frac{8}{15}(4-y)^{\frac{5}{2}} \right]_0^4$$

Fully integrated

(A1)

$$I = \frac{256}{15}$$

Correct limit applied to their integral

(A1F)

$$\bar{y} = \frac{I}{32/3} = \frac{8}{5}$$

Their evaluated  $I \div$  area

(m1)(A1)

**General for all three versions:**

M1 Attempt to integrate an appropriate  
function of  $x$  or  $y$  (must apply a full method)

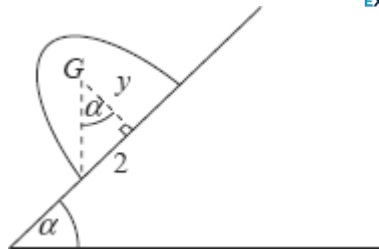
A1 Correct integration

A1F Correct limits applied to their integral

m1 Their evaluated integral  $\div$  area

A1 Correct answer  $\frac{8}{5}$

(b)



$$\tan \alpha = \frac{2}{y}$$

*tan alpha seen*

M1

*Correct structure – ft error in (a)*

A1F

$$\tan \alpha = \frac{2}{8/5} = 1.25$$

*Substitute and use of  $\tan^{-1}$  – dependent on first M1*

m1

$$\alpha = 51^\circ$$

*ft error in (a)*

A1F

4

[9]

EXAM PAPERS PRACTICE

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$$(a) \int_0^{2r} xy^2 \, dx = \int_0^{2r} \frac{x^3}{4} \, dx$$

*Attempt to use formula  $\int xy^2 \, dx$*

M1

$$= \left[ \frac{x^4}{16} \right]_0^{2r}$$

$$= r^4$$

*Integration correct*

A1

$$\int_0^{2r} y^2 dx = \int_0^{2r} \frac{x^2}{4} dx$$

Or use of  $\frac{1}{3} \pi r^2 h$  to get  $\frac{2}{3} \pi r^3$

$$= \left[ \frac{x^3}{12} \right]_0^{2r}$$

$$= \frac{2r^3}{3}$$

B1

$$\Rightarrow \bar{x} = r^4 \div \frac{2r^3}{3} = \frac{3r}{2}$$

$$\bar{x} = \frac{\pi \int_0^{2r} xy^2 dx}{\pi \int_0^{2r} y^2 dx}$$

AG use of

NB – consistent use of  $\pi$  throughout for M1A1 at end (or cancelled at start)

M1A1

5

(b) (i)

|       | mass                            | distance           |
|-------|---------------------------------|--------------------|
| Lower | $\pi r^2 (2r) \rho$             | $r$                |
| Upper | $\frac{\pi r^3}{3} (2r) k \rho$ | $2r + \frac{r}{2}$ |

Any correct pairing seen anywhere  
(mass  $\leftrightarrow$  distance)

B1

$$\left( \pi 2r^3 \rho + \frac{\pi 2r^3}{3} k \rho \right) \bar{x} = \pi 2r^3 \rho (r)$$

Equation formed

M1

$$+ \frac{\pi 2r^3}{3} k \rho \left( \frac{5r}{2} \right)$$

lose 1 each 'type' of error

$$\Rightarrow \left(1 + \frac{k}{3}\right) \bar{x} = r + \frac{5rk}{6}$$

$$\Rightarrow (6 + 2k) \bar{x} = (6 + 5k) r$$

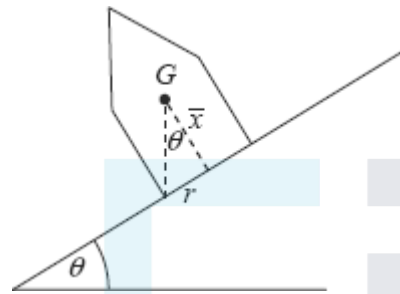
$$\bar{x} = \left(\frac{6 + 5k}{6 + 2k}\right) r$$

Rearrange to obtain printed answer

A1

5

(ii)



$$\tan \theta = \frac{r}{\bar{x}}$$

Use of  $\tan \theta$

Correct structure

M1

A1

$$\Rightarrow \frac{2}{3} = \frac{r}{\left(\frac{6 + 5k}{6 + 2k}\right) r}$$

Substitution of  $\bar{x}$ ,  $\tan \theta$

B1

$$\frac{2}{3} = \frac{6 + 2k}{6 + 5k}$$

$$12 + 10k = 18 + 6k$$

Attempt to solve

M1

$$4k = 6$$

$$k = \frac{3}{2}$$

A1

5

**Q8.**

- (a) On point of toppling, take moments about bottom right corner

*attempt at moments*

M1

$$W(2a) = P \cos \theta (8a)$$

*A1 each side*

A1,A1

$$P = \frac{W}{4 \cos \theta}$$

A1

4

- (b) On point of sliding vertically,  $N + P \sin \theta = W$

M1A1

horizontally,  $F = P \cos \theta$

M1A1

friction  $F = \mu N$

$$\Rightarrow P \cos \theta = \mu(W - P \sin \theta)$$

*substitute; use of  $F = \mu N$*

M1A1

$$P \cos \theta = \mu W - \mu P \sin \theta$$

$$P(\cos \theta + \mu \sin \theta) = \mu W$$

$$P = \frac{\mu W}{\cos \theta + \mu \sin \theta}$$

*AG*

A1

7

- (c) Slides before topples  $\Rightarrow$

$$\frac{\mu W}{\cos \theta + \mu \sin \theta} < \frac{W}{4 \cos \theta}$$

*inequality formed*

M1

$$4\mu \cos \theta < \cos \theta + \mu \sin \theta$$

*elimination of fractions / cancel W*

A1

$$4\mu < 1 + \mu \tan \theta$$

*÷ by  $\cos \theta$  and use of  $\tan \theta = 1$*

A1

$$\tan \theta = 1 \Rightarrow 3\mu < 1$$

*collect  $\mu$  terms*

M1

$$\mu < \frac{1}{3}$$

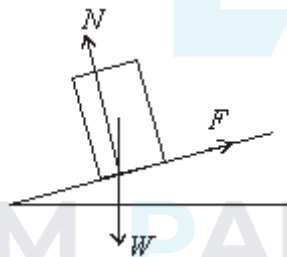
A1

5

[16]

**Q9.**

(a)



*Three forces*

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B1

**Alternative:** *N/F combined*

*Through one point*

B1

2

(b) About to topple  $\Rightarrow \tan \alpha = \frac{d}{3d}$

*Use of  $\tan \alpha$*

$$\frac{d}{3d} \text{ used or } \frac{3d}{d}$$

M1

$$\therefore \alpha = \tan^{-1} \left( \frac{1}{3} \right)$$

$$= 18.4^\circ$$

*Accept 18°*

A1

3

(c) Parallel to plane

$$W \sin \alpha = F$$

*Attempt to resolve parallel or perpendicular to the plane*

M1

Perpendicular to plane

$$W \cos \alpha = N$$

*Both correct*

A1

Law of friction

$$F \leq \mu N$$

$$\mu = \frac{2}{9} \therefore W \sin \alpha \leq \frac{2}{9} W \cos \alpha$$

*Law of friction used*

M1

$$\tan \alpha \leq \frac{2}{9}$$

*$\alpha$  tan or  $\alpha$  obtained or inequality stated*

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A1

$$\frac{2}{9} < \frac{1}{3} \text{ or } 13^\circ < 18^\circ$$

Slide first

*Comparison and conclusion – correct answer only  
[N.B. Accept (b) and (c) in any order]*

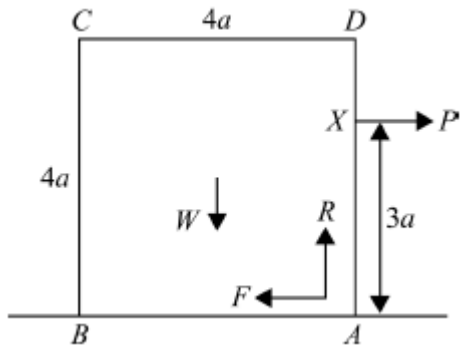
A1

5

[10]

**Q10.**





- (a) Block sliding  $\therefore P > F$

M1

$$R = W$$

B1

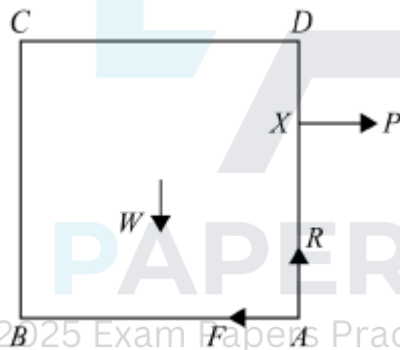
$$F = \mu R$$

hence  $P > \mu W$

A1

3

- (b) (i)



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forces

A1

1

(ii)  $P \times 3a \geq W \times 2a$

M1

$$\Rightarrow P \geq \frac{2W}{3} \quad \text{allow = for A1}$$

A1

2

- (c) If  $\mu = 0.6$ , slides when  $P > 0.6W$ , if not toppled previously

B1

If  $\mu = 0.667$ , topples when  $P > 0.667W$  if not started to slide

B1

$\therefore$  slides first

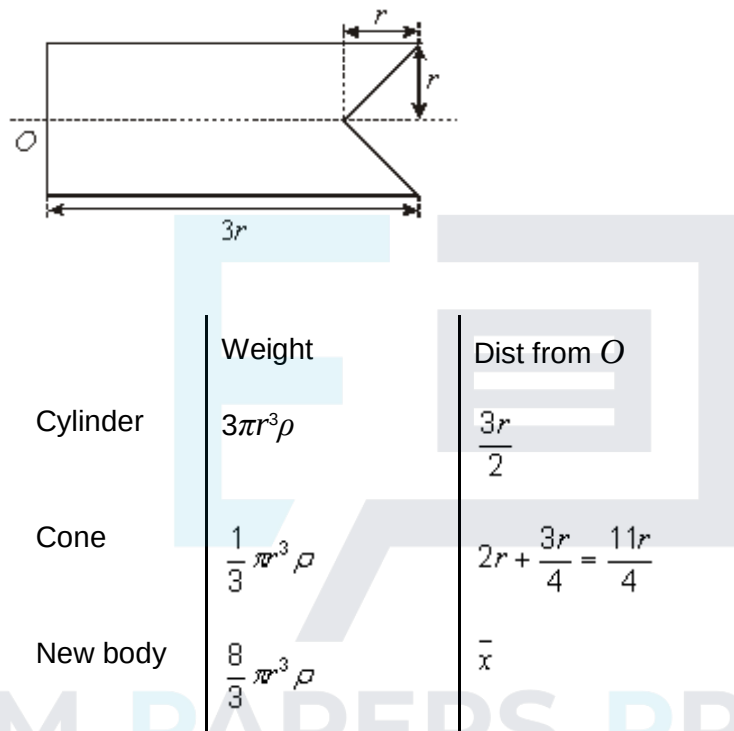
B1

3

[9]

**Q11.**

(a)



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M1

-1EE

A2

$$\frac{8}{3} \pi r^3 \rho \bar{x} + \frac{1}{3} \pi r^3 \rho \times \frac{11r}{4} = 3\pi r^3 \rho \times \frac{3r}{2}$$

*correct combination*

M1

*at least 2 weight  $\times$  dist correct*

A1

$$\frac{8\bar{x}}{3} = \frac{9r}{2} - \frac{11r}{12}$$

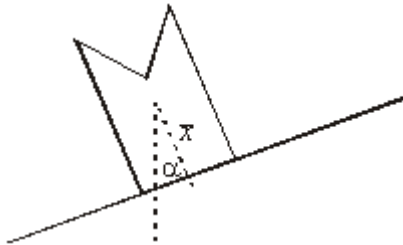
$$\Rightarrow 32\bar{x} = 54r - 11r$$

$$\Rightarrow \bar{x} = \frac{43r}{32}$$

A1

6

(b)



Body will topple when  $\alpha > \tan^{-1} \frac{r}{\bar{x}}$

M1

$$\alpha > \frac{\tan^{-1} r}{\frac{43r}{32}}$$

A1f.t.

$$> \tan^{-1} \frac{32}{43}$$

$$> 36.7^\circ$$

A1f.t.

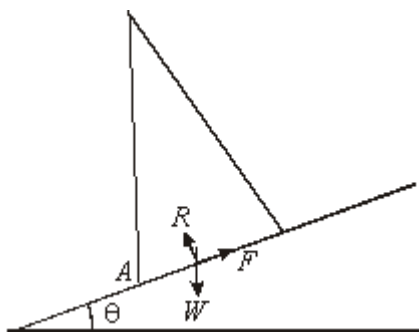
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3

[9]

**Q12.**

(a)



for  $R, F, W$

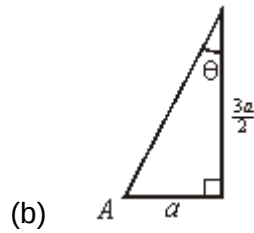
B1

for concurrence shown

B0 if weight shown starting from centre of base

B1

2



C.G. is  $\frac{1}{4} \times 6a$  from base =  $\frac{3a}{2}$

B1

cone topples if  $\theta > \frac{\tan^{-1} a}{\frac{3a}{2}}$   
 for understanding line of action of  $W$  passes through  $A$ . May be shown on a diagram

M1

$> \tan^{-1} \frac{2}{3} (= 33.7^\circ)$

A1

3

(c) either  $R = W \cos \theta$   
 or solve

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$F = W \sin \theta$   
 both correct

A1

$F \leq \mu R$

M1

$\therefore W \sin \theta \leq \mu W \cos \theta$

A1

$\Rightarrow \tan \theta \leq 0.6$   
 $\theta \leq 31.0^\circ$

A1

$$(or \tan^{-1} 0.6 < \tan^{-1} \frac{2}{3})$$

∴ The cone will slide before it topples  
*ft on c's  $\theta$  if calculated*

E1F

6

*Alternative:*

$$F = W \sin \theta$$

M1 A1

$$R = W \cos \theta$$

A1

$$\text{slips when } \frac{F}{R} = \mu$$

M1

$$\text{i.e. } \tan \theta = 0.6$$

A1

$$< \frac{2}{3}$$

*slides first*

E1

(6)

[11]

**Q13.**

- (a) At point of sliding  
 Vertically;  $R = Mg - P \sin \theta$

M1 A1

$$\text{Horizontally; } F = P \cos \theta$$

M1 A1

$$F = \frac{1}{5} R$$

B1

$$\frac{1}{5} Mg - \frac{1}{5} P \sin \theta = P \cos \theta$$

$$P = \frac{Mg}{5 \cos \theta + \sin \theta}$$

A1

6

- (b) At point of toppling  
Taking moments about A  
 $Mgl = P \cos \theta \, l$

*For moments about A and one side correct*

M1

A1

$$P = \frac{Mg}{7 \cos \theta}$$

A1

3

- (c) If topples before it slides

$$\frac{Mg}{7 \cos \theta} < \frac{Mg}{5 \cos \theta + \sin \theta}$$

M1

$$Mg (5 + \tan \theta) < 7 Mg$$

M1

A1✓

$$5 + \tan \theta < 7$$

*Use of  $\leq$  M3 A1*

M1

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$$\tan \theta < 2$$

*Use of  $>$  M2 A1*

A1

5

[14]

**Q14.**

$$\bar{x} = \frac{\int_0^5 x(0.4x)^2 dx}{\int_0^5 (0.4x)^2 dx}$$

(a)

*Use of appropriate expression for  $\bar{x}$*

M1

*Correct expression excluding limits*

A1

Correct limits of integration

A1

$$\begin{aligned}
 &= \frac{\int_0^5 x^3 dx}{\int_0^5 x^2 dx} \\
 &= \frac{625/4}{125/3} = \frac{15}{4}
 \end{aligned}$$

Evaluation of integrals

m1

**ag** Correct result from correct working using both limits

A1

5

(b) Radius = 2

Radius of face = 2

B1

$$\tan \alpha = \frac{2}{5/4} = \frac{8}{5}$$

Use of tan

M1

Use of  $5 - \bar{x}$

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M1

Correct expression for  $\tan \alpha$

A1

$$\alpha = \tan^{-1}\left(\frac{8}{5}\right) = 58.0^\circ$$

Correct angle; allow  $58^\circ$

A1

5

[10]

**Q15.**

- (a) If  $\bar{x}$  is the distance of the centre of mass from the base  
Moments about plane of the base

M1

$$\pi r^2 \cdot 2r \cdot \rho + \frac{1}{3} \pi r^2 H (2r + \frac{H}{4}) \cdot 3\rho$$

M1A1

$$= (\pi r^2 \cdot 2r \rho + \frac{1}{3} \pi r^2 H \cdot 3\rho) \bar{x}$$

A1

$$2r^2 + 2Hr + \frac{H^2}{4} = (2r + H) \bar{x}$$

Dep both M1

m1

$$\bar{x} = \frac{8r^2 + 8Hr + H^2}{4(2r + H)}$$

A1

6

(b) If on the point of toppling

$$\tan \theta = \frac{1}{5} = \frac{r}{\bar{x}}$$

$$\therefore 5r = \frac{8r^2 + 8Hr + H^2}{4(2r + H)}$$

for  $\frac{r}{5}$  or  $5r =$  result in (a)

M1

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A1

$$40r^2 + 20rH = 8r^2 + 8rH + H^2$$

M1

$$32r^2 + 12rH - H^2 = 0$$

$$H^2 - 12rH - 32r^2 = 0$$

A1

$$H = \frac{12 \pm \sqrt{144 + 128}}{2} r$$

$$= (6 \pm 2\sqrt{17}) r$$

$$= (6 + 2\sqrt{17}) r$$



**Q16.**

- (a) Moments about A

$$P \cdot 4l \cos \alpha$$

*M1 awarded even for moments about A even when on horizontal floor or if P. 4l seen*

M1 A1

$$= mg(l \cos \alpha - 2l \sin \alpha)$$

A1

$$P = \frac{\cos \alpha - 2 \sin \alpha}{4 \cos \alpha} mg$$

*m instead of mg used penalise one A1 in question*

A1

4

- (b) Resolve along the plane

$$F - P \cos \alpha = mg \sin \alpha$$

M1 A1

Resolve perpendicular to the plane

$$P \sin \alpha + R = mg \cos \alpha$$

M1 A1

Using  $F = \mu R$

B1

$$mg \sin \alpha + P \cos \alpha = \mu (mg \cos \alpha - P \sin \alpha)$$

$$P \cos \alpha + \mu P \sin \alpha = \mu mg \cos \alpha - mg \sin \alpha$$

M1

$$P = \frac{\mu \cos \alpha - \sin \alpha}{\cos \alpha + \mu \sin \alpha} mg$$

$$\text{Accept } P = \frac{\mu - \tan \alpha}{1 + \mu \tan \alpha} mg$$

A1

7

**Q17.**

Mass of cylinder is  $\pi(r)^2 l \cdot \rho = \pi r^2 l \rho$

*Condone not using  $\rho$  as long as 6 used*

B1

Mass of cuboid is  $(2r)^2 \cdot r \cdot 6\rho$

B1

$$= 24r^3 \rho$$

Taking moments about A (at point of toppling)

M1

$$24r^3 \rho \left( \frac{r}{2} \sin \alpha - r \cos \alpha \right) = 0$$

M1A1

$$+ \pi r^2 l \rho \left\{ \left( \frac{1}{2} + r \right) \sin \alpha - r \cos \alpha \right\} = 0$$

M1A1

$$12r^2 \tan \alpha - 24r^2 + \pi \frac{l^2}{2} \tan \alpha \dots$$

$$\dots + \pi r l \tan \alpha - \pi r l = 0$$

M1

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$$\pi l^2 - 6\pi r l - 168 r^2 = 0$$

A1

9

**Or using centre of gravity**

Mass of cylinder is  $\pi(r)^2 l \cdot \rho = \pi r^2 l \rho$

B1

Mass of cuboid is  $(2r)^2 \cdot r \cdot 6\rho = 24r^3 \rho$

B1

Let  $\bar{x}$  be distance of C of G from base of trophy

Taking moments about plane of base

$$24r^3 \rho \frac{r}{2} + \pi r^2 l \rho \left( \frac{1}{2} + r \right)$$

M1

$$= \{24r^3 \rho + \pi r^2 l \rho\} \bar{x}$$

A1

$$12r^2 + \frac{\pi}{2} l^2 + \pi r l = \{24r + \pi l\} \bar{x}$$

M1

$$\bar{x} = \frac{12r^2 + \frac{\pi}{2} l^2 + \pi r l}{24r + \pi l}$$

A1

If, on point of toppling, C of G is vertically above A

M1

$$\bar{x} \sin \alpha = r \cos \alpha$$

$$(24r^2 + 2\pi r l + \pi l^2) \tan \alpha = 48r^2 + 2\pi r l$$

M1

$$6r^2 + \frac{1}{2} \pi r l + \frac{1}{4} \pi l^2 = 48r^2 + 2\pi r l$$

$$\pi l^2 - 6\pi r l + 168 r^2 = 0$$

M1

EXAM PAPERS PRACTICE [9]

**Q18.** ©2025 Exam Papers Practice. All rights reserved.

(a) Resolve vertically  $R = Mg - P \sin \theta$

M1 A1

Resolve horizontally  $F = P \cos \theta$

B1

On point of sliding  $F = \mu R$

B1

$$\mu(Mg - P \sin \theta) = P \cos \theta$$

M1

$$P = \frac{\mu Mg}{\cos \theta + \mu \sin \theta}$$

A1

6

(b) Taking moments about A

M1

$$5l \times P \cos \theta = l \times W$$

A1

$$P = \frac{Mg}{5 \cos \theta}$$

B1

3

(c) If slides before it topples,

$$\frac{\mu Mg}{\cos \theta + \mu \sin \theta} < \frac{Mg}{5 \cos \theta}$$

M1

$$5\mu \cos \theta < \cos \theta + \mu \sin \theta$$

M1

$$5\mu < 1 + \mu \tan \theta$$

A1

$$\tan \theta = \frac{1}{7}, 5\mu < 1 + \frac{1}{7}\mu$$

M1

$$\mu < \frac{7}{34}$$

A1

5

[14]

## Examiner reports

### Q2.

This question was well answered. Almost all students chose to find the limiting  $P$  value for each case and then set up the inequality by arguing which value of  $P$  would be reached first. To score full marks, students had to clearly state why the inequality was needed by referring to sliding occurring before toppling. On occasions the inequality was reversed and not corrected. A more significant error was to use the mass of 0.4 kg when taking moments, leaving out  $g$ .

### Q3.

There was an improved response on this topic than in the past as candidates have clearly learnt the required formula. Integration was generally successful. Some candidates chose to ignore the given volume and recalculated it. The use of  $\pi$  was generally consistent although occasionally got left in the final answer. Many candidates scored full marks on this part. Part (b)(i) was well done and where candidates did not split  $P$  into components they did correctly use a compound trig expression which was expanded correctly using an identity from core 4. Part (b)(ii) and (iii) proved more challenging but a significant number of candidates scored full marks here too.

### Q4.

The majority of candidates were aware of the required formulae for part (a). Candidates should always state the full formulae first before cancelling the  $\pi$ . In part (b)(i) candidates were aware of the three forces but failed to have the forces all passing through one point. In part (b)(ii) most candidates scored at least two marks; quite often the  $r/d$  ratio was incorrect. Part (c) was well understood, but candidates needed to make an explicit comparison for full marks to be awarded.

### Q5.

This was a question in which candidates scored very well indeed. In part (a), using  $P \sin \theta$  was a common error. Almost all candidates obtained full marks on part (b). In part (c), the inequality was sometimes reversed. Candidates' reasoning in part (d) was excellent.

### Q6.

It was disappointing to see a significant number of candidates make heavy weather of this question. Candidates should be aware of the formulae stated in the formulae booklet. A number of candidates started from first principles and needed to do a lot of work to score the first mark. Many who chose this approach could not complete the integration, particularly if they had left the integrand as a function of  $y$ . Numerous errors occurred with incorrect limits. Part (b) was done well and added to candidates' scores.

### Q7.

There was a very pleasing response to this question. All marks were lost in part (a) if a two-dimensional formula was used. The best solutions in part (b) used a tabular approach before setting up the relevant equation. A common error was to have the centre of mass of the cone at a distance of  $3.5r$  from the base, instead of  $2.5r$ . The last part was well understood with the correct principle applied. The best responses included a clear labelled diagram showing the principle concerned.

### Q8.

Topic – Toppling and sliding. There was an excellent response to this question. Part (a) was almost universally correct; however when slips did occur it was because of the use of  $W(4a)$  when taking moments.

Part (b) was rarely seen incorrect, however inequality signs sometimes lost a mark. Part (c) was answered very well with sound knowledge of cancelling, and trigonometrical identities. Once again a mark was sometimes lost because of an equality or greater than sign.

### Q9.

Only a few candidates correctly drew the force diagram, many not realising the **three** forces must go through the same point for the second mark. Many candidates found that the structure here did not help them with many answering part (b) in part (c). Full marks were awarded for this, of course. When errors did occur it was often due to an inverted ratio in part (b) – 3 instead of  $1/3$ .

### Q10.

This question required clarity of thought. Most candidates were able to make some progress. The use of the word “toppling”, as in the specification, led some candidates mistakenly to draw a diagram with the block rotated through  $45^\circ$ .

### Q11.

This was a situation where the final body was obtained by subtracting the cone from the cylinder, and this caused candidates difficulties in drawing up an appropriate “moments equation”. With the answer to the first part having been given, the solution to the final “toppling” problem was standard of its type. Candidates should be aware that they are not penalised here if they have not satisfactorily answered the first part of such a question, but go on to use the information given to pick up later marks.

### Q12.

Many candidates failed to gain the mark awarded for showing clearly the concurrence property of three forces in equilibrium. A number showed the weight of the cone acting as if from the centre of the base, rather than the centre of gravity. Parts (b) and (c) required clear thinking and logical argument. This proved difficult for the weaker candidates.

### Q13.

Candidates answered this question well, with only a small number needing creative algebra to obtain the answer given in part (a).

Part (b) was also well answered.

In part (c), a few used incorrect inequalities but the majority started with

$\frac{Mg}{7 \cos \theta} < \frac{Mg}{5 \cos \theta + \sin \theta}$ . The algebraic manipulation needed to obtain  $\tan \theta < 2$  was found difficult by significant numbers of candidates. A surprising number simplified the correct initial inequality to  $7 \cos \theta < 5 \cos \theta + \sin \theta$ .

### Q14.

In part (a), the candidates tended to score either zero marks or full marks. The candidates who knew which approach to take were generally able to complete their

$$\bar{x} = \frac{\int xy dx}{\int y dx}$$

solutions. A number of candidates used  $\int xy dx$ . Some candidates made errors in their integration and some others made errors when squaring  $0.4x$ , typically obtaining  $0.4x^2$ . Part (b) was much more demanding with very few correct answers, but a lot of candidates picked up one or two marks on this part of the question. The two main issues were that candidates did not find the radius of the cone and also that they did not find the distance of the centre of mass from the base of the cone.

### Q15.

Better candidates made good progress with this question. In part (a) problems encountered by other candidates included finding the volume of a cone, in which the

third in the formula  $\frac{1}{3}\pi r^2 h$  was omitted or was incorrect; the mass of the cone, in which the density  $3\rho$  was not used; and finding the position of the centre of mass of the cone. Part (b) was answered well by many candidates.

### Q16.

Although the diagram printed in the question paper clearly showed the cuboid to be on an inclined plane, a significant proportion of candidates redrew the diagram so that the cuboid rested on a horizontal plane. In part (a), most appreciated that the cuboid would topple about  $A$ , but some assumed that the weight of the cuboid would thus act through  $A$ , ignoring the force  $P$ . Moments about  $A$  often produced correctly  $P \cos \alpha$ ,  $4l$ , but the moment of the weight was often incorrect.

Part (b) of the question was generally done well and, although it was relatively rare to see full equations (for example, resolving along the plane), the correct methods were implied and the correct value of  $P$  was usually seen.

### Q17.

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This question proved to be challenging. A number of candidates could not use the densities given to find the masses of the cylinder and the cuboid. Most appreciated that the centre of gravity was vertically above the lowest edge when the trophy was about to topple. The common correct method was to find the distance of the centre of

gravity above the base,  $\bar{x} = \frac{12r^2 + \frac{\pi}{2}l^2 + \pi l}{24r + \pi l}$ , and then to use  $\bar{x} \sin \alpha = r \cos \alpha$  at the point of toppling.

### Q18.

This question was well done by a majority of candidates. Part (b) was usually correct; only a few tried to pivot the cuboid, which introduced extra angles.

In part (c), some had the inequality the wrong way round. Other simplified

$\frac{\mu Mg}{\cos \theta + \mu \sin \theta}$  into  $\frac{\mu Mg}{\cos \theta} + \frac{\mu Mg}{\mu \sin \theta}$  but most candidates obtained  $\frac{7}{34}$  as required.

A few obtained 0.206 which was accepted; the equivalent answer to two significant figures was not accepted.



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